

BDS-3/GPS uncombined real-time PPP with PPP-B2b precise products: Modeling and its long-term performance evaluation

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Abstract

Through PPP-B2b real-time precise products, BDS-3 can provide the real-time precise point positioning (PPP) service without relying on the Internet for users in China and its surrounding regions. In this study, the BDS-3/GPS uncombined real-time PPP model with PPP-B2b precise products is constructed. In the constructed model, a receiver clock offset parameter is introduced for each satellite system to account for the frequent changes of GPS satellite clock datums, and a compensation parameter is also introduced for each satellite to mitigate the effects of initial systematic bias of satellite clock offsets. In addition, the long-term positioning performance of BDS-3/GPS real-time PPP with PPP-B2b precise products is evaluated using the observations from six stations spanning 196 days. The results show that the average convergence time can be 7.5 and 10.2 min (with a standard deviation (STD) of 3.9 and 6.4 min) over the analysis period in static and kinematic modes, respectively, which is shortened by 41.3 % and 61.4 % compared with the traditional model that ignores the influence of frequently switched datums and initial systematic bias of satellite clock offsets. The average positioning accuracy can be 4.6, 0.9, and 2.5 cm (with a STD of 1.3, 0.7, and 1.7 cm), and 6.4, 3.3, and 9.5 cm (with a STD of 2.3, 1.5, and 3.1 cm) in east, north, and up directions in static and kinematic modes, respectively. These results indicate that the BDS-3/GPS real-time PPP with PPP-B2b precise products can provide stable and reliable real-time high-precision navigation and positioning service.

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Keywords: BDS-3 PPP-B2b; Uncombined model; Satellite clock datums; Initial systematic bias; Real-time precise point positioning; Long-term performance

1. Introduction

Precise Point Positioning (PPP) is an absolute positioning technique that uses code and carrier phase observations from a Global Navigation Satellite System (GNSS) receiver, along with precise satellite orbit and clock products generated by the International GNSS Service (IGS). PPP

can determine the three-dimensional (3D) coordinates within the International Terrestrial Reference Framework (ITRF) through comprehensively considering and correcting various errors (Malys and Jensen, 1990; Zumberge et al., 1997; Kouba and Héroux, 2001). However, the real-time PPP (RT-PPP) typically requires IGS precise products in real time via the Internet, known as the Real-Time Service (RTS) (Guo et al. 2024). Consequently, it cannot achieve real-time high-precision positioning without the Internet access.

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As one of the services provided by BDS-3, the precise positioning service (i.e., PPP-B2b) is a kind of open and free high-precision service provided for users by the BDS-3 Geostationary Orbit (GEO) satellites, which broadcast precise satellite orbit, clock offset, and differential code bias (DCB) corrections to users in China and its surrounding areas (Yang et al. 2022). Because PPP-B2b service is broadcast through GEO satellites, users can achieve real-time high-precision positioning without the Internet access. Currently, the PPP-B2b service is widely used in some fields such as integrated navigation (Xiao et al. 2023), earthquake source inversion (Zang et al. 2024), time transfer (Ge et al. 2023) and so on.

In PPP data processing, the precise satellite orbit and clock products are indispensable. Additionally, the DCB products are also crucial to improve the positioning performance of PPP. Currently, the PPP-B2b service only provides corrections for BDS-3 and GPS satellites, but may provide the corrections of other GNSS constellations in the future. The IGS precise products are generated using hundreds of GNSS ground tracking stations distributed globally (Dash et al. 2024), and the accuracy of PPP-B2b products is worse than that of IGS final products due to limitations in the distribution and quantity of reference stations (Tang et al. 2022). Benefitting from the inter-satellite link (ISL), the accuracy of along-track component for the BDS-3 orbit corrections in PPP-B2b is better than that of GPS ones, but the corresponding accuracy of radial and cross-track components is comparable for them. However, the PPP-B2b orbit corrections are not as accurate as IGS RTS products (Tao et al. 2021). Regarding the satellite clocks, the accuracy of clock corrections in PPP-B2b is slightly worse than that of rapid products from GeoForschungsZentrum (GFZ) (Zhang et al. 2022). Considering the combined effects of PPP-B2b orbit and clock errors, the residual Signal-in-Space Ranging Error (SISRE) can be estimated to improve the performance of PPP-B2b RT-PPP (Xu et al. 2023). In addition, the DCB in PPP-B2b is consistent with the DCB from IGS precise products and Time Group Delay (TGD) from broadcast ephemerides. The difference of DCB between PPP-B2b and IGS precise products is less than 2 ns, and the difference between PPP-B2b DCB and TGD is less than 0.15 ns (Ren et al. 2021).

In recent years, there have been a lot of research on the positioning performance of PPP-B2b. China Satellite Navigation Office (CSNO) states in its official document that the PPP-B2b service of BDS-3 aims to provide users in China and its surrounding areas with positioning accuracies (with a confidence level of 95 %) of better than 30 and 60 cm in the horizontal and vertical directions for the BDS-3 single system, and of better than 20 and 40 cm in the horizontal and vertical directions for the BDS-3/GPS dual-system after full convergence (CSNO 2021). Xu et al. (2021) found that BDS-3 dual-frequency kinematic RT-PPP can achieve the accuracy of 11 and 17 cm in the horizontal and vertical directions after conver-

gence, respectively, and the B1C/B2a ionospheric-free (IF) combination showed better positioning accuracy than the B1I/B3I IF combination. Geng et al. (2022) reported that the RT-PPP with PPP-B2b service could achieve the root mean square (RMS) statistics of 3D positioning errors of 21.3 cm for BDS-only case and 18.2 cm for GPS/BDS integrated case in coastal areas, respectively. Zhou et al. (2023) discovered that the positioning accuracies of dual-frequency static and kinematic RT-PPP based on PPP-B2b service were better than 10 and 20 cm, respectively. Wu et al. (2023) discovered that when using PPP-B2b corrections, the positioning errors with a confidence level of 95 % for static stations, low-speed objects (such as vehicles and ships) and high-kinematic airborne platforms using BDS-3 single-system and BDS-3/GPS dual-system combination were all less than 20 and 35 cm in the horizontal and vertical directions after convergence, respectively. These studies demonstrate that PPP-B2b service can provide decimeter-level positioning accuracy in the kinematic mode.

Most of the previous research about PPP-B2b employed only a few days of observation data (the maximum duration of the used data is one month) to evaluate RT-PPP performance, which makes it difficult to accurately represent the real-time positioning performance of PPP-B2b service and provide effective advice for BDS-3 users. Moreover, most of the existed studies on PPP-B2b focused on dual-frequency IF combined model, and the research on undifferenced and uncombined (UDUC) multi-system integrated RT-PPP still remains insufficient, especially in which the various biases have not been systematically investigated. Given the prevalence of UDUC PPP, this paper aims to construct a BDS-3/GPS UDUC RT-PPP model based on PPP-B2b service, considering the frequently switched datums and the included initial systematic bias of PPP-B2b satellite clock offsets, in order to improve the positioning accuracy and accelerate the convergence speed of PPP-B2b RT-PPP. This paper is divided into the following parts. We first introduce the precise corrections broadcast by PPP-B2b service. Then, a BDS-3/GPS UDUC RT-PPP model based on PPP-B2b service is constructed. Next, we present and discuss the results. Finally, we summarize the conclusions.

2. Precise corrections broadcast by PPP-B2b service

According to Interface Control Document (ICD) of PPP-B2b (CSNO 2020), PPP-B2b service only broadcasts BDS-3 and GPS corrections. As shown in Table 1, there are four types of corrections that users can receive and decode: satellite mask, satellite orbit correction and user range accuracy index, DCB, and satellite clock correction, with updating intervals of 48, 48, 48, and 6 s, respectively. For the corrections beyond these intervals, the data quality cannot be guaranteed. According to ICD, the PPP-B2b information is used to correct the CNAV-1 navigation messages of BDS-3 B1C signal and the LNAV navigation mes-

Table 1
PPP-B2b information types.

Type	Type name	Updating interval (s)	Validate time (s)	BDS-3	GPS
1	Satellite mask	48	–	✓	✓
2	Orbit correction and user range accuracy index	48	96	✓	✓
3	DCB	48	86,400	✓	–
4	Satellite clock correction	6	12	✓	✓

sages of GPS L1/L2 signals, respectively. BDS-3 CNAV-1 uses an 18-parameter model for orbit calculation, while GPS L-NAV uses a 16-parameter model. It is worth noting that BDS-3 adopts the BDS Time (BDT) as the time reference when using PPP-B2b precise corrections and conducting RT-PPP data processing. During preprocessing, it is necessary to correct the 14-second offset with respect to GPS Time (GPST). Meanwhile, BDS-3 adopts the BeiDou Coordinate System (BDCS), whose definition complies with the standards of the latest ITRF frame.

The satellite mask contains 255 bits. When the bit is “1”, it represents that the PPP-B2b information of the corresponding satellite is broadcast. In addition, because the other three corrections of PPP-B2b are designed for IF combinations, it cannot be directly applied to the UDUC RT-PPP model. Therefore, the methods of using PPP-B2b information based on the UDUC model are provided as follows.

2.1. Orbit correction

The orbit corrections in the PPP-B2b include the components of the orbit correction vector δO in the radial, along-track, and cross-track (RAC) directions. The orbit corrections are used to calculate the satellite position correction vector δX , which is then combined with the satellite position vector $X_{broadcast}$ calculated from the broadcast ephemerides to obtain the precise satellite position vector X_{orbit} , that is:

$$X_{orbit} = X_{broadcast} - \delta X \quad (1)$$

where X_{orbit} denotes the precise satellite position corrected by the orbit corrections, $X_{broadcast}$ is the satellite position calculated by the broadcast ephemerides, and δX represents the satellite orbit corrections, whose calculation method can refer to ICD of PPP-B2b.

It is worth noting that the reference of PPP-B2b orbit corrections is different from the widely used reference of IGS precise orbit products. For PPP-B2b, BDS-3 orbit corrections refer to the antenna phase center (APC) of B3I, while GPS orbit corrections refer to the APC of L1/L2 IF combination. However, IGS precise orbit products refer to the center of mass (CoM). Therefore, in UDUC RT-PPP processing, it is necessary to apply the antenna phase center offset (PCO) and variation (PCV) corrections from the ANTenna Exchange format (ANTEX) file (igs20.atx file) to the calculated satellite positions and multi-frequency

observations. To be consistent, the reference of satellite positions is firstly transferred to the CoM, and then when using the observations of a certain frequency, the reference is transferred to the APC on that frequency. The correspondence between APC and CoM can be expressed as follows (Montenbruck et al. 2015):

$$\begin{cases} r_{CoM}^G = r_{APC}^G - A_G^T \cdot \left(\frac{(f_1^G)^2}{(f_1^G)^2 - (f_2^G)^2} a_{PCO,1}^G - \frac{(f_2^G)^2}{(f_1^G)^2 - (f_2^G)^2} a_{PCO,2}^G \right) \\ r_{CoM}^C = r_{APC}^C - A_C^T \cdot a_{PCO,3}^C \\ r_{APC,j}^G = r_{CoM}^G + A_G^T \cdot a_{PCO,j}^G \\ r_{APC,j}^C = r_{CoM}^C + A_C^T \cdot a_{PCO,j}^C \end{cases} \quad (2)$$

where j denotes the frequency, C represents BDS-3, G represents GPS, r_{APC} and r_{CoM} denote the satellite position of APC (of L1/L2 IF combination for GPS, and of B3I signal for BDS-3, respectively) and CoM, respectively, $r_{APC,j}^G$ denotes the satellite position of APC on frequency j , A_C^T and A_G^T represent the attitude matrix of BDS-3 and GPS satellites, respectively, f_1^G and f_2^G denote the carrier frequency of L1 and L2 signals, respectively, $a_{PCO,1}^G$ and $a_{PCO,2}^G$ denote the GPS PCO corrections of L1 and L2 signals, respectively, and $a_{PCO,3}^C$ represents the BDS-3 PCO corrections of B3I signal.

2.2. Clock correction

The clock corrections in PPP-B2b are provided as the correction parameter relative to the broadcast satellite clocks. The application of these corrections can be expressed as:

$$t_{satellite} = t_{broadcast} - \frac{C_0}{c} \quad (3)$$

where $t_{broadcast}$ refers to the satellite clock offset calculated by the broadcast ephemerides, $t_{satellite}$ denotes the precise satellite clock offset corrected by clock corrections, c is the speed of light in vacuum, and C_0 represents the clock corrections acquired from PPP-B2b message. It is noted that GPS clock offsets refer to L1/L2 IF combination, while BDS-3 clock offsets refer to B3I signal. As a result, the calculated precise satellite clock of GPS inevitably assimilates the satellite code hardware delay of L1/L2 IF combination, while BDS-3 clock offset assimilates the satellite code hardware delay of B3I signal.

2.3. DCB correction

In view that the satellite code hardware delay included in the single-frequency code observations is inconsistent with the assimilated code hardware delay in the PPP-B2b precise satellite clock offsets, it is necessary to apply DCB corrections to the code observations to uniform the datum, that is:

$$\tilde{l}_{sig} = l_{sig} - DCB_{sig} \quad (4)$$

where $\tilde{l}_{sig}$ denotes the corrected code observations of signal sig , l_{sig} represents the raw code observations, and DCB_{sig} denotes the DCB corrections. For BDS-3, the satellite clock offset refers to B3I signal, and DCB also refers to B3I signal. If users adopt other signals, the DCB corrections broadcast in the PPP-B2b message can be directly applied to correct the code observations.

However, the PPP-B2b message does not contain the DCB corrections of GPS, and it is recommended that users utilize the L1/L2 dual-frequency IF combined observations. This is because the GPS clock offsets derived from PPP-B2b message refer to L1/L2 IF combination, and contain the satellite code hardware delay of L1/L2 IF combination. When using the calculated GPS clock offsets, the GPS code hardware delay at the satellite will be eliminated simultaneously. Actually, for dual-frequency UDUC RT-PPP, the satellite code hardware delay will be assimilated by the ionospheric delay parameters when DCB corrections are not applied, and thus the positioning results are not affected.

3. BDS-3/GPS UDUC RT-PPP model based on PPP-B2b service

PPP-B2b service uses GNSS reference stations distributed throughout China to track satellites for observation, and calculates precise satellite orbit, clock offset, and DCB corrections through ground operation control systems. The derived corrections have regional and real-time characteristics. The real-time precise satellite clock offsets typically comprise three parts (Mervart and Weber 2011):

$$Clk_a^s = O_a + O_a^s + Clk^s \quad (5)$$

where s represents satellites, a denotes analysis centers, O_a represents the datum of satellite clock offsets, which is the same for all satellites, O_a^s represents the initial systematic bias introduced by initial satellite clock offsets, which is a constant within a continuous arc, and Clk^s denotes the actual satellite clock offsets. The magnitude of initial systematic bias O_a^s in the IGS final precise clock offset products is much small, and thus it can be neglected in the data processing. However, for satellite clock offsets in PPP-B2b, the influence of O_a^s cannot be ignored.

Because the recovered GPS and BDS-3 satellite clocks with PPP-B2b message refer to the L1/L2 IF combination and B3I signal, respectively, they will inevitably absorb the corresponding code hardware delay at the satellite. The formula can be expressed as:

$$\begin{cases} Clk_a^{s,G} = O_a^G + O_a^{s,G} + Clk^{s,G} - (\alpha_{12}^G \cdot d_1^{s,G} + \beta_{12}^G \cdot d_2^{s,G}) \\ Clk_a^{s,C} = O_a^C + O_a^{s,C} + Clk^{s,C} - d_3^{s,C} \end{cases} \quad (6)$$

with $\alpha_{12}^G = (f_1^G)^2 / [(f_1^G)^2 - (f_2^G)^2]$ and $\beta_{12}^G = -(f_2^G)^2 / [(f_1^G)^2 - (f_2^G)^2]$ where C refers to BDS-3, G refers to GPS, α_{12}^G and β_{12}^G are the dual-frequency IF combination coefficients, f_1^G and f_2^G denote the carrier frequency of L1 and L2 signals, respectively, $d_1^{s,G}$ and $d_2^{s,G}$ are GPS satellite code hardware delay of L1 and L2 signals, respectively, and $d_3^{s,C}$ is BDS-3 satellite code hardware delay of B3I signal.

In GNSS RT-PPP, for a satellite observed by a receiver, there are two types of observations, which can be expressed as follows:

$$\begin{cases} P_{r,j}^{s,G} = \rho_r^{s,G} + cdt_r^G - cdt^{s,G} + T_r^{s,G} + \mu_j^G I_{r,1}^{s,G} + d_{r,j}^G + d_j^{s,G} + \varepsilon(P_{r,j}^{s,G}) \\ L_{r,j}^{s,G} = \rho_r^{s,G} + cdt_r^G - cdt^{s,G} + T_r^{s,G} - \mu_j^G I_{r,1}^{s,G} + N_{r,j}^{s,G} + b_{r,j}^G + b_j^{s,G} + \varepsilon(L_{r,j}^{s,G}) \\ P_{r,j}^{s,C} = \rho_r^{s,C} + cdt_r^C - cdt^{s,C} + T_r^{s,C} + \mu_j^C I_{r,1}^{s,C} + d_{r,j}^C + d_j^{s,C} + \varepsilon(P_{r,j}^{s,C}) \\ L_{r,j}^{s,C} = \rho_r^{s,C} + cdt_r^C - cdt^{s,C} + T_r^{s,C} - \mu_j^C I_{r,1}^{s,C} + N_{r,j}^{s,C} + b_{r,j}^C + b_j^{s,C} + \varepsilon(L_{r,j}^{s,C}) \end{cases} \quad (7)$$

where P and L represent code and carrier phase observations in meters, respectively, s and r denote a GNSS satellite and a receiver, respectively, j denotes the frequency (j with a numerical value of 1 and 2 refers to L1 and L2 for GPS, respectively, while j with a numerical value of 1 and 3 refers to B1I and B3I for BDS-3, respectively), C denotes BDS-3, G denotes GPS, ρ

($\rho = \sqrt{(x_r - x^s)^2 + (y_r - y^s)^2 + (z_r - z^s)^2}$) is the geometric distance from the receiver antenna phase center to the satellite antenna phase center, (x_r, y_r, z_r) and (x^s, y^s, z^s) represent the 3D coordinates of the receiver and satellite in the Earth-Centered Earth-Fixed (ECEF) frame, respectively, cdt_r and cdt^s denote receiver clock offset and satellite clock offset, respectively, T is the slant tropospheric delay, μ_j ($\mu_j = f_1^2 / f_j^2$) is the ionospheric delay factor on frequency j , f is the carrier frequency, I_1 is the slant ionospheric delay on the first frequency, N is the phase ambiguity in units of meters, d_r and d^s denote the receiver and satellite code hardware delay, respectively, b_r and b^s denote the receiver and satellite phase hardware delay, respectively, and ε represents the observation noise including multipath effects.

When applying the corrections of PPP-B2b message, the satellite clock datum, initial systematic bias and satellite code hardware delay included in the satellite clock offsets (see Equation (6)) will be introduced into the observation

equations. Therefore, after linearizing the raw observations, the widely used observation equations can be expressed as:

$$\begin{cases} p_{r,j}^{s,G} = \mu_r^{s,G} \cdot x + cdt_r^{-G} + m_r^{s,G} \cdot Z_r + \mu_j^G \bar{I}_{r,1}^{-s,G} + O_a^{s,G} \\ l_{r,j}^{s,G} = \mu_r^{s,G} \cdot x + cdt_r^{-G} + m_r^{s,G} \cdot Z_r - \mu_j^G \bar{I}_{r,1}^{-s,G} + \bar{N}_{r,j}^{-s,G} + O_a^{s,G} \\ p_{r,j}^{s,C} = \mu_r^{s,C} \cdot x + cdt_r^{-G} + ISB_r^{C,G} + m_r^{s,C} \cdot Z_r + \mu_j^C \bar{I}_{r,1}^{-s,C} + O_a^{s,C} \\ l_{r,j}^{s,C} = \mu_r^{s,C} \cdot x + cdt_r^{-G} + ISB_r^{C,G} + m_r^{s,C} \cdot Z_r - \mu_j^C \bar{I}_{r,1}^{-s,C} + \bar{N}_{r,j}^{-s,C} + O_a^{s,C} \end{cases} \quad (8)$$

with

$$\begin{cases} cdt_r^{-G} = cdt_r^G + \alpha_{12}^G d_{r,1}^G + \beta_{12}^G d_{r,2}^G + O_a^G \\ ISB_r^{C,G} = (cdt_r^C - cdt_r^G) + (\alpha_{13}^C d_{r,1}^C + \beta_{13}^C d_{r,3}^C) - (\alpha_{12}^G d_{r,1}^G + \beta_{12}^G d_{r,2}^G) + (O_a^C - O_a^G) \\ \bar{I}_{r,1}^{-s,G} = I_{r,1}^{s,G} + \beta_{12}^G (d_{r,1}^G - d_{r,2}^G) + \beta_{12}^G (d_{r,1}^{s,G} - d_{r,2}^{s,G}) \\ \bar{I}_{r,1}^{-s,C} = I_{r,1}^{s,C} + \beta_{13}^C (d_{r,1}^C - d_{r,3}^C) \\ \bar{N}_{r,j}^{-s,G} = N_{r,j}^{s,G} + b_{r,j}^G + b_j^{s,G} - (\alpha_{12}^G - \mu_j^G \beta_{12}^G) (d_{r,1}^G + d_{r,2}^{s,G}) - \beta_{12}^G (1 + \mu_j^G) (d_{r,2}^G + d_{r,2}^{s,G}) \\ \bar{N}_{r,j}^{-s,C} = N_{r,j}^{s,C} + b_{r,j}^C + b_j^{s,C} - (\alpha_{13}^C - \mu_j^C \beta_{13}^C) d_{r,1}^C - \beta_{13}^C (1 + \mu_j^C) d_{r,3}^C - d_{r,3}^{s,C} \end{cases} \quad (9)$$

with $\alpha_{13}^C = (f_1^C)^2 / [(f_1^C)^2 - (f_3^C)^2]$ and $\beta_{13}^C = -(f_3^C)^2 / [(f_1^C)^2 - (f_3^C)^2]$ where p and l denote the observed-minus-computed (OMC) observables of code and carrier phase, respectively, μ is the unit vector from the receiver to the satellite; x refers to the position vector of the receiver, cdt_r denotes the receiver clock offset that assimilates the receiver code hardware delay and satellite clock datum of GPS, ISB denotes the inter-system bias (ISB) between BDS-3 and GPS, which is the sum of inter-system difference of time scale, dual-frequency IF combined receiver code hardware delay and satellite clock datum between BDS-3 and GPS, m represents the wet mapping function, Z is the tropospheric zenith wet delay (ZWD) at the station,

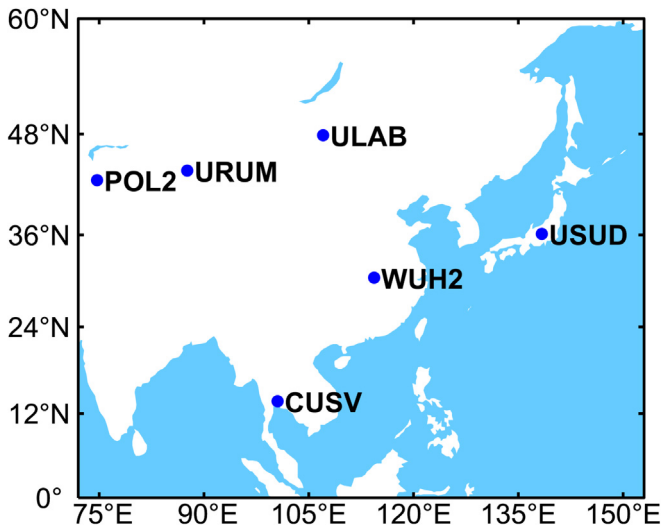


Fig.1. Distribution of six selected MGEX stations.

$\bar{I}_1$ represents the ionosphere delay that absorbs the code hardware delay at the receiver for BDS-3 and at both satellite and receiver for GPS, $\bar{N}$ denotes the phase ambiguity that absorbs the phase and code hardware delay at both satellite and receiver. Other error items, such as PCO, PCV, tropospheric dry delay (Saastamoinen 1972), phase wind-up effect (Wu et al. 1993), relativistic effect (Ashby 2003), station displacement and so on, are assumed to be corrected by precise products or empirical models.

The satellite clock datum is introduced to eliminate the rank deficiency between receiver and satellite clock offsets during the real-time precise clock estimation (PCE), and is the common item for all satellite clock offsets within a system. As shown in Equation (9), during the RT-PPP processing, the satellite clock datum of a selected system is absorbed by the receiver clock parameter, and the difference of satellite clock datum between two systems is absorbed by the ISB parameter. Thus, the satellite clock datums introduced by satellite clock offsets do not affect the position solutions (Yao et al. 2017). However, the satellite clock datum of GPS satellite clock offsets in the PPP-B2b changes frequently, and the ISB parameter (usually modeled as random-walk process) cannot well absorb the difference of satellite clock datum between BDS-3 and GPS. As a result, the positioning results will be negatively affected. Therefore, it is necessary to estimate the receiver clock offset parameters (usually modeled as white noise process) separately for the BDS-3 and GPS systems. As such, the satellite clock datums of BDS-3 and GPS satellite clock offsets will be absorbed by the respective receiver clock offset parameters. When applying PPP-B2b clock offset products, it is necessary to make appropriate modifications at the traditional UDUC RT-PPP observation model, that is:

$$\begin{cases} p_{r,j}^{s,G} = \mu_r^{s,G} \cdot x + cdt_r^{-G} + m_r^{s,G} \cdot Z_r + \mu_j^G \bar{I}_{r,1}^{-s,G} + O_a^{s,G} \\ l_{r,j}^{s,G} = \mu_r^{s,G} \cdot x + cdt_r^{-G} + m_r^{s,G} \cdot Z_r - \mu_j^G \bar{I}_{r,1}^{-s,G} + \bar{N}_{r,j}^{-s,G} + O_a^{s,G} \\ p_{r,j}^{s,C} = \mu_r^{s,C} \cdot x + cdt_r^{-C} + m_r^{s,C} \cdot Z_r + \mu_j^C \bar{I}_{r,1}^{-s,C} + O_a^{s,C} \\ l_{r,j}^{s,C} = \mu_r^{s,C} \cdot x + cdt_r^{-C} + m_r^{s,C} \cdot Z_r - \mu_j^C \bar{I}_{r,1}^{-s,C} + \bar{N}_{r,j}^{-s,C} + O_a^{s,C} \end{cases} \quad (10)$$

with

$$\begin{cases} cdt_r^{-G} = cdt_r^G + \alpha_{12}^G d_{r,1}^G + \beta_{12}^G d_{r,2}^G + O_a^G \\ cdt_r^{-C} = cdt_r^C + \alpha_{13}^C d_{r,1}^C + \beta_{13}^C d_{r,3}^C + O_a^C \end{cases} \quad (11)$$

where cdt_r^{-G} and cdt_r^{-C} denote the GPS and BDS-3 receiver clock offset, respectively.

During the PPP-B2b real-time PCE, when a satellite becomes visible in the Asian-Pacific region, an absolute satellite clock offset should be first introduced. However, the error of the introduced absolute satellite clock offset will result in the initial systematic bias O_a^s in the real-time

Table 2
Processing strategies of RT-PPP.

Items	Processing strategies
Frequency	BDS-3: B1I, B3I; GPS: L1, L2
Elevation cut-off angle	10°
Sampling interval	30 s
Sampling	Elevation-dependent weights
Estimator	Kalman filter
Satellite orbit	PPP-B2b products
Satellite clock offset	PPP-B2b products
DCB	PPP-B2b products
Ionospheric delay	Estimated as random-walk process
Tropospheric delay	ZWD is estimated as random-walk process, and Global Mapping Function (GMF) is used as wet mapping function
Receiver clock offset	Estimated as white noise process
Phase ambiguity	Estimated as float constants
Cycle slip detection	Geometry Free and Melbourne–Wübbena combinations
Satellite PCO/PCV	Corrected using IGS20.atx file
Station coordinate	Estimated as constants in the static mode, and white noise process in the kinematic mode

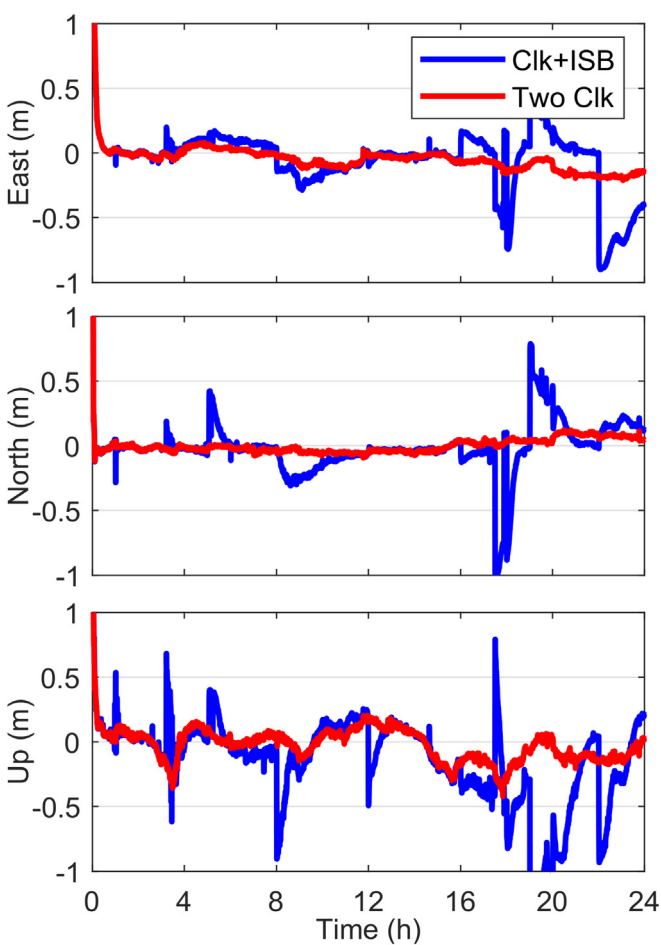


Fig. 2. RT-PPP positioning errors at POL2 station for “Clk + ISB” and “Two Clk” models.

precise satellite clock offsets (Zhang et al. 2011). In the carrier phase observation equation, this bias will be absorbed by the phase ambiguity parameter to form a new ambiguity parameter, which has no effect on the solutions of other parameters (Chen et al. 2017). However, for code observa-

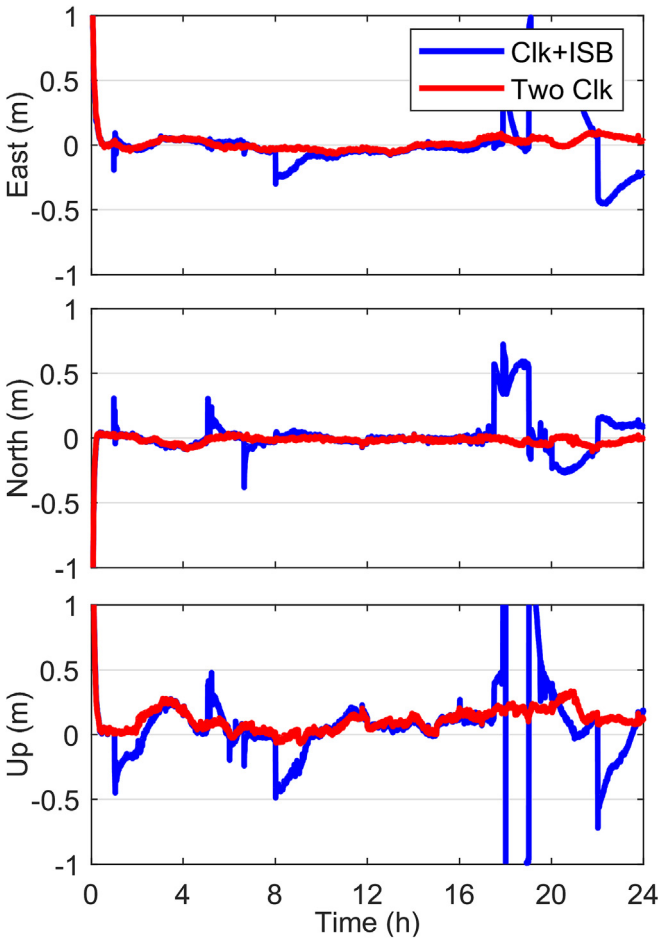


Fig. 3. RT-PPP positioning errors at ULAB station for “Clk + ISB” and “Two Clk” models.

tions, this bias will reduce their accuracy. At the early stage of RT-PPP processing, the positioning accuracy is primarily determined by code observations, and their low accuracy will increase the convergence time of position solutions. To mitigate the influence of initial systematic

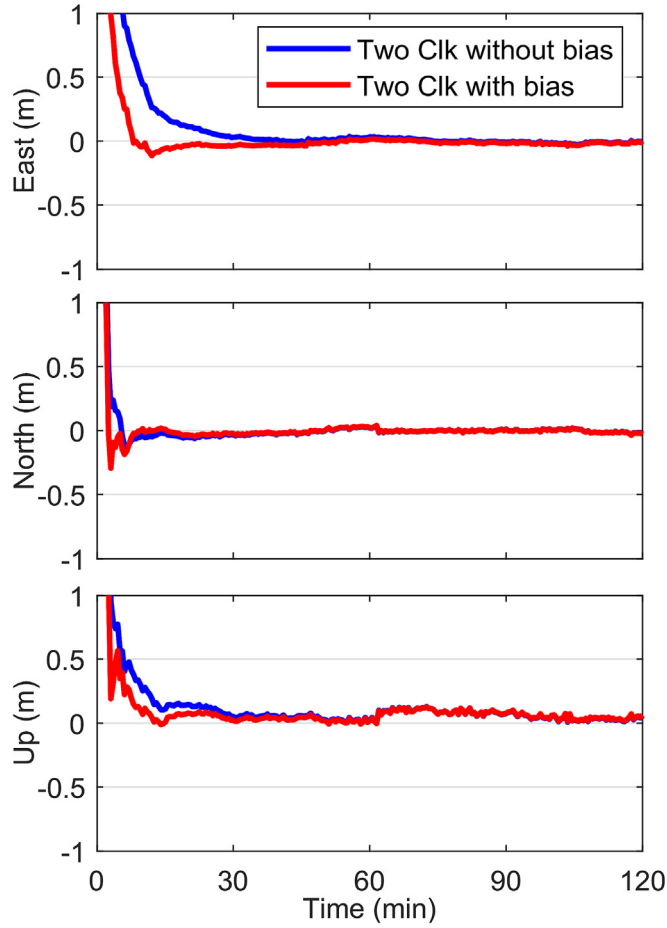


Fig. 4. RT-PPP positioning errors at POL2 station for “Two Clk with bias” and “Two Clk without bias” models.

bias, a compensation parameter needs to be set for the code observation equation of each satellite. The initial systematic bias is a constant within a continuous arc, but will change significantly when the satellite is tracked again. At this time, the compensation parameter corresponding to the satellite should be initialized. Considering the effect of satellite clock datum and initial systematic bias of real-time satellite clock offsets, the final BDS-3/GPS UDUC RT-PPP model based on PPP-B2b can be described as:

$$\begin{cases} p_{r,j}^{s,G} = \mu_r^{s,G} \cdot x + cdt_r^{-G} + m_r^{s,G} \cdot Z_r + \mu_j^G I_{r,1}^{-s,G} + \sigma^{s,G} \\ l_{r,j}^{s,G} = \mu_r^{s,G} \cdot x + cdt_r^{-G} + m_r^{s,G} \cdot Z_r - \mu_j^G I_{r,1}^{-s,G} + \tilde{N}_{r,j}^{s,G} \\ p_{r,j}^{s,C} = \mu_r^{s,C} \cdot x + cdt_r^{-C} + m_r^{s,C} \cdot Z_r + \mu_j^C I_{r,1}^{-s,C} + \sigma^{s,C} \\ l_{r,j}^{s,C} = \mu_r^{s,C} \cdot x + cdt_r^{-C} + m_r^{s,C} \cdot Z_r - \mu_j^C I_{r,1}^{-s,C} + \tilde{N}_{r,j}^{s,C} \end{cases} \quad (12)$$

with

$$\begin{cases} \sigma^{s,G} = O_a^{s,G} \\ \sigma^{s,C} = O_a^{s,C} \\ \tilde{N}_{r,j}^{s,G} = \tilde{N}_{r,j}^{-s,G} + O_a^{s,G} \\ \tilde{N}_{r,j}^{s,C} = \tilde{N}_{r,j}^{-s,C} + O_a^{s,C} \end{cases} \quad (13)$$

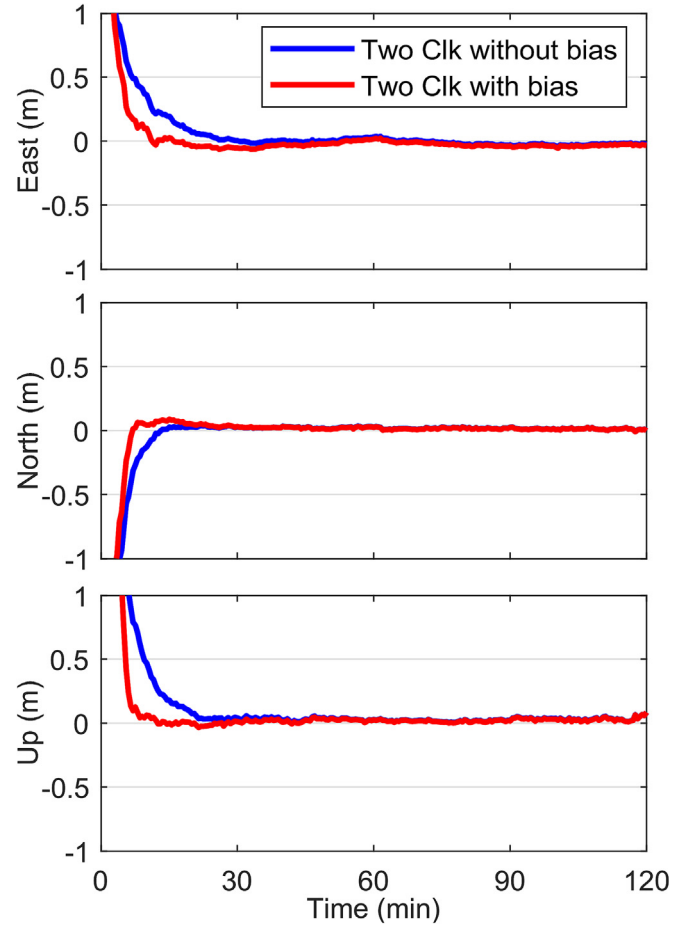


Fig. 5. RT-PPP positioning errors at ULAB station for “Two Clk with bias” and “Two Clk without bias” models.

where σ denotes the compensation parameter of initial systematic bias, which is modeled as random-walk process, and $\tilde{N}$ denotes the new phase ambiguity grouped with the initial systematic bias. To eliminate the rank deficiency of this model, a virtual observation equation that the compensation parameter of a selected satellite is equal to zero needs to be introduced for each system.

4. Results and discussion

4.1. Experimental data and processing strategy

In order to assess the positioning performance of the constructed BDS-3/GPS UDUC RT-PPP model based on PPP-B2b service, we choose six Multi-GNSS Experiment (MEGX) multi-frequency and multi-system stations for corresponding experiments. The analysis period covers a total of 196 days from day of year (DOY) 251 in 2023 to DOY 81 in 2024. Fig. 1 shows the distribution of the selected stations. Considering that all three BDS-3 satellites broadcasting PPP-B2b messages are GEO satellites with sub-satellite point of 80°E, 110.5°E and 114°E in longitude, all the selected stations are located in China and its surrounding area. Thus, these stations can receive PPP-B2b

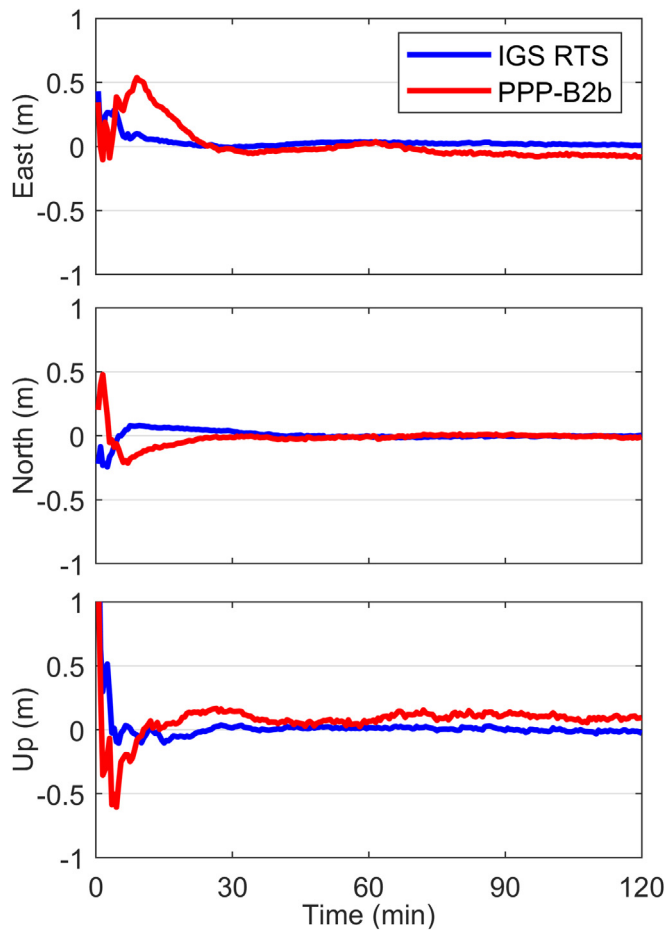


Fig. 6. RT-PPP positioning errors at CUSV station based on IGS RTS and PPP-B2b service.

messages and track sufficient BDS-3 and GPS satellites. And the PPP-B2b products are decoded through the receiver developed by Changsha Jinwei Integrated Circuit Co., Ltd. The self-developed software is used in this paper, which also draws on some open software, such as RTKLIB software (Takasu 2013). And the processing strategies of RT-PPP are shown in Table 2.

4.2. Impact of frequently switched satellite clock datum

With the PPP-B2b message, the BDS-3/GPS UDUC RT-PPP model constructed in this paper sets one clock offset parameter for each system (hereinafter referred to as the “Two Clk” model) to account for the frequent changes in GPS satellite clock datum. This is compared to the traditional RT-PPP model, which sets one receiver clock offset parameter and one ISB parameter (hereinafter referred to as the “Clk + ISB” model).

Figs. 2 and 3 show the RT-PPP positioning errors for “Clk + ISB” and “Two Clk” models at POL2 and ULAB stations on DOY 251 of 2023 respectively. The kinematic mode is employed. The RMS statistics of positioning errors in east, north, up, and 3D directions at POL2 and ULAB stations under the “Clk + ISB” model are 30.6, 20.4,

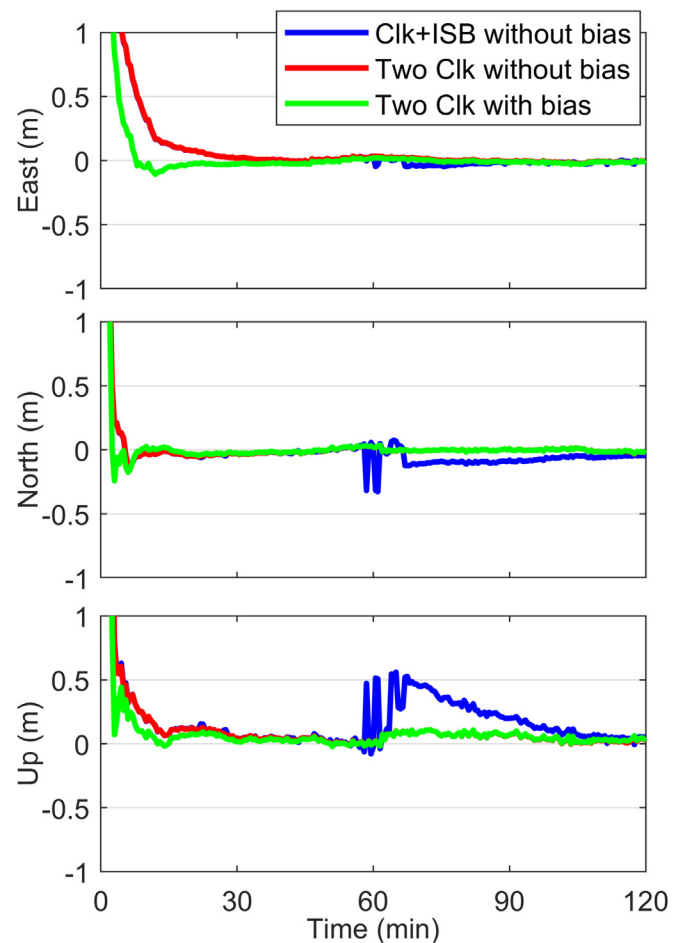


Fig. 7. IF RT-PPP positioning errors at POL2 station for “Clk + ISB without bias”, “Two Clk without bias”, and “Two Clk with bias” models.

55.4, and 66.5 cm, and 24.5, 17.1, 46.5, and 55.2 cm, respectively. Under the “Two Clk” model, the corresponding RMS statistics are reduced by 32.1 %, 65.9 %, 28.5 %, and 31.9 %, and 64.1 %, 51.5 %, 48.5 %, and 51.5 % to 20.8, 7.0, 39.7, and 45.3 cm, and 8.8, 8.3, 23.9, and 26.8 cm, respectively. The figures show that the positioning accuracies in the three directions under two different models are comparable after full convergence. However, due to some changes in the GPS satellite clock datum, the positioning accuracy of the “CLK + ISB” model suddenly becomes worse many times even after full convergence, making the positioning performance unstable and unreliable. In contrast, the “Two Clk” model effectively absorbs the changes in the GPS satellite clock datum, providing more stable positioning accuracy without sudden changes.

4.3. Impact of initial systematic bias in satellite clock offset

With the PPP-B2b messages, the BDS-3/GPS UDUC RT-PPP model constructed in this paper sets one compensation parameter for each satellite (hereinafter referred to as the “Two Clk with bias” model) to mitigate the negative impact of initial systematic bias in satellite clock offset on

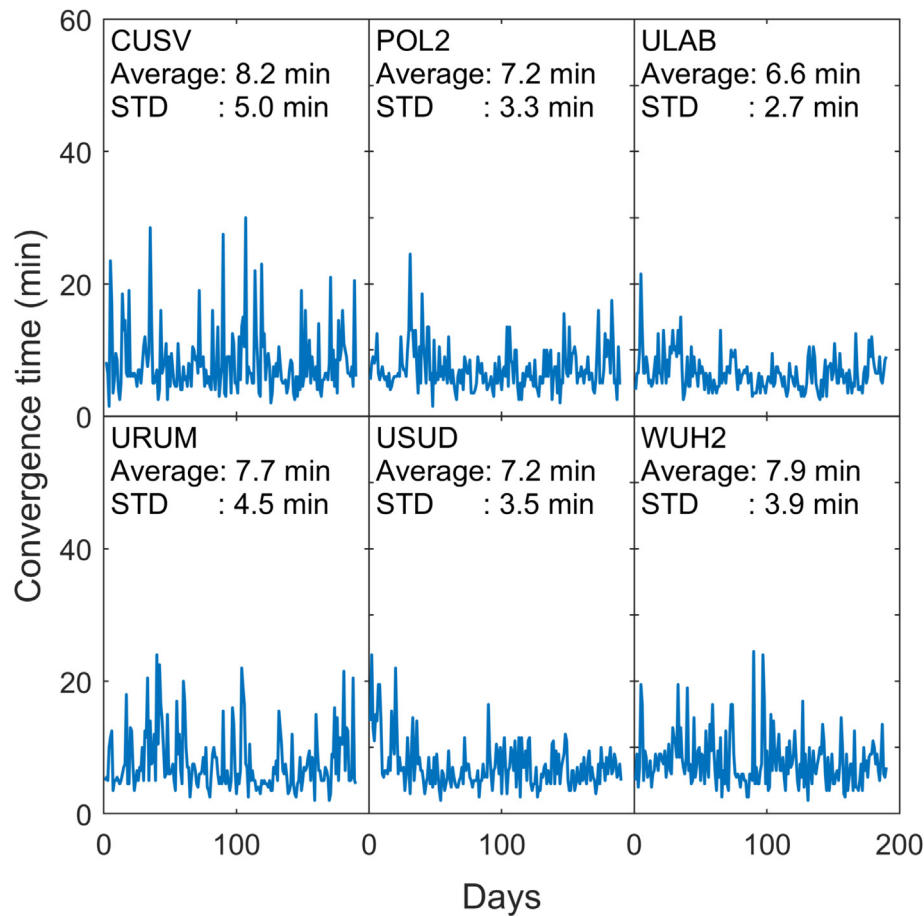


Fig. 8. Daily convergence time in 196 days at six stations in the static mode.

the RT-PPP positioning performance. This is compared to the traditional model without the compensation parameter for the initial systematic bias (hereinafter referred to as the “Two Clk without bias” model).

In this study, the position solution is considered to have converged when the positioning errors remain below 20 cm in the horizontal direction and below 40 cm in the vertical direction for at least five consecutive minutes. Figs. 4 and 5 illustrate the RT-PPP positioning errors for “Two Clk with bias” and “Two Clk without bias” models at POL2 and ULAB stations in the first two hours on DOY 251 of 2023 respectively. The kinematic mode is employed. The figures show that the convergence time is significantly reduced after introducing the compensation parameter, which is shortened by 53.6 % and 52.0 % from 14.0 and 12.5 min to 6.5 and 6.0 min at POL2 and ULAB stations, respectively. After full convergence, the positioning accuracy with the compensation parameter is comparable to that without the compensation parameter.

4.4. Comparison with IGS RTS and assessment of IF model

IGS has been providing real-time services for over ten years. And the Centre National d’Etudes Spatiales (CNES) provides archival precise orbit/clock products, based on which UDUC RT-PPP can be implemented. Fig. 6 shows

the positioning errors based on IGS RTS and PPP-B2b service at CUSV station in the first two hours on DOY 251 of 2023. The kinematic mode is employed. As shown in the figure, the positioning performance based on IGS RTS is better than that based on PPP-B2b service, particularly at the early stage of positioning. Furthermore, the observation data of the six selected stations on DOY 251 of 2023 are processed to analyze positioning performance, and all the parameters are reset every two hours. According to the positioning results over six stations, the average convergence time of PPP-B2b and IGS RTS in the kinematic mode is 11.9 and 9.8 min, respectively, demonstrating that the RT-PPP positioning performance based on IGS RTS is better than that based on PPP-B2b.

According to the UDUC model constructed above in this paper, three BDS-3/GPS IF RT-PPP models (“Clk + ISB without bias”, “Two Clk without bias”, and “Two Clk with bias”) are constructed. Fig. 7 shows the RT-PPP positioning errors for “Clk + ISB without bias”, “Two Clk without bias”, and “Two Clk with bias” models at POL2 station in the first two hours on DOY 251 of 2023. The kinematic mode is employed. The figure demonstrates that similar with UDUC RT-PPP model, setting a receiver clock offset parameter for each satellite system can reduce the sudden changes of positioning errors, making it more stable. Meanwhile, setting one compensation parameter

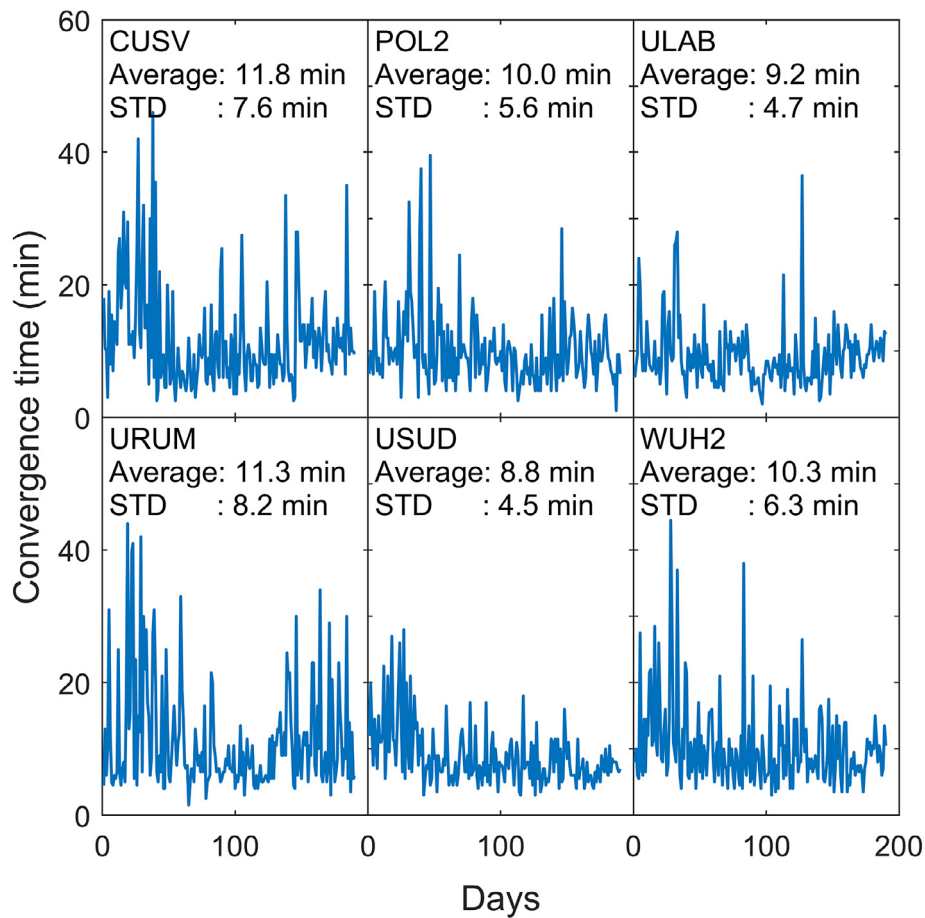


Fig. 9. Daily convergence time in 196 days at six stations in the kinematic mode.

Table 3

Average and STD statistics of BDS-3/GPS UDUC RT-PPP positioning accuracy in the east/north/up/3D directions in the static and kinematic modes.

Station	Static (cm)		Kinematic (cm)	
	Average	STD	Average	STD
CUSV	4.7/0.7/1.8/5.3	2.3/0.6/1.4/2.2	8.1/3.4/11.4/14.7	3.4/1.6/3.4/4.2
POL2	5.5/1.6/2.5/6.6	1.2/1.1/2.2/1.7	7.7/4.5/10.2/13.8	2.3/1.8/4.0/4.1
ULAB	4.0/0.6/2.9/5.2	1.0/0.4/1.3/0.9	4.4/2.7/8.9/10.4	1.6/0.9/1.9/2.1
URUM	5.8/1.0/2.6/6.7	1.2/0.7/1.8/1.3	7.4/3.4/8.9/12.2	2.1/1.3/2.7/2.8
USUD	3.4/1.3/2.4/4.6	1.1/1.0/1.9/1.9	4.4/3.0/8.6/10.2	2.1/1.4/3.7/4.2
WUH2	4.0/0.4/2.6/5.0	1.0/0.3/1.3/1.1	6.6/2.8/8.7/11.5	2.6/2.2/2.9/4.0

for each satellite can accelerate convergence speed and significantly shorten convergence time. In comparison with “Two Clk without bias” model, the convergence time of “Two Clk with bias” model is reduced by 43.5 % from 11.5 to 6.5 min.

4.5. Assessment of long-term RT-PPP positioning performance

Based on the BDS-3/GPS UDUC RT-PPP model constructed in this paper (namely “Two Clk with bias” model) and the traditional RT-PPP model ignoring the frequently

switched datum and the included initial systematic bias of PPP-B2b satellite clock offsets (namely “Clk + ISB without bias” model), this section evaluates the long-term positioning performance using the observations from DOY 251 in 2023 to DOY 81 in 2024 at six stations, in terms of convergence time and positioning accuracy.

With the BDS-3/GPS UDUC RT-PPP model constructed in this paper, Figs. 8 and 9 provide the daily convergence time in 196 days at six stations in the static and kinematic modes, respectively. The average and standard deviation (STD) statistics of convergence time over 196 days are computed for each station, and they are pro-

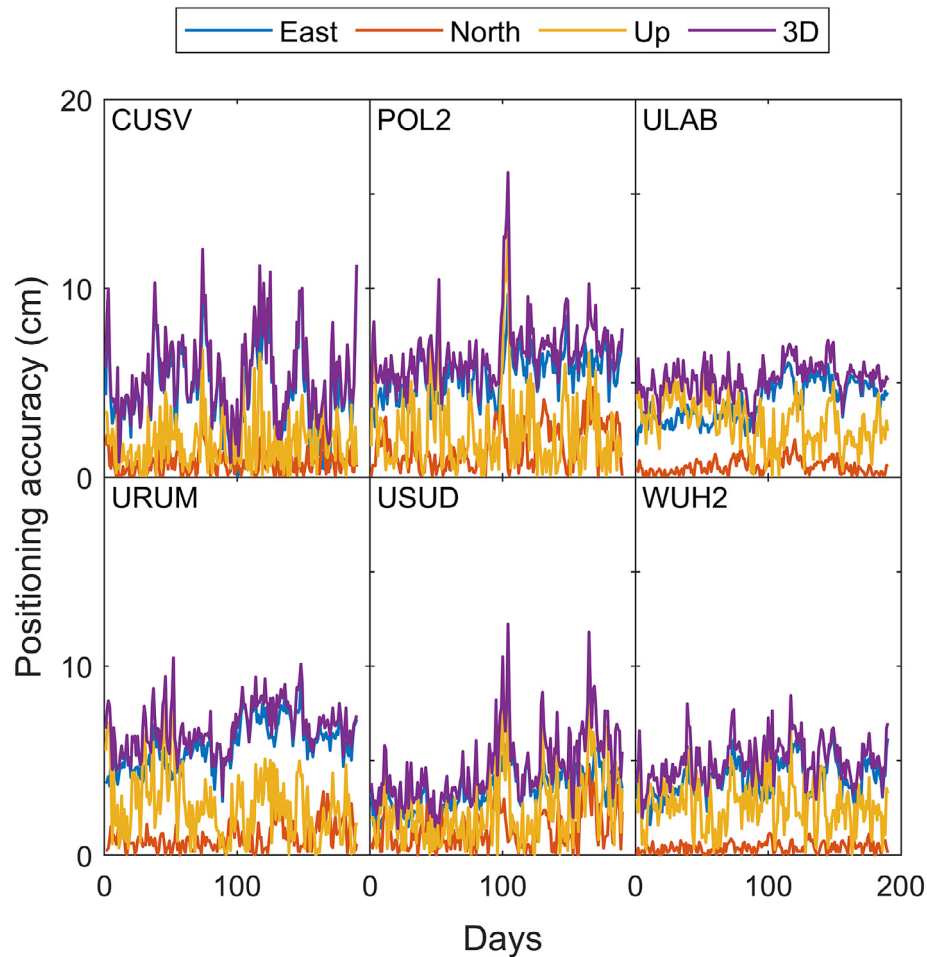


Fig. 10. Daily positioning accuracy in 196 days at six stations in the static mode.

vided in each panel. According to the average results over six stations, the average convergence time is 7.5 and 10.2 min in the static and kinematic modes, respectively, which is shortened by 41.3 % and 61.4 % over the traditional model with the corresponding statistics being 12.7 and 26.5 min, respectively. This demonstrates that the PPP-B2b UDUC RT-PPP model constructed in this paper can achieve fast convergence of position solutions within a much short period of time. Moreover, during the experiment period, the STD of convergence time is 3.9 and 6.4 min in the static and kinematic modes, respectively. From perspective of convergence time, this model can provide users with stable and reliable positioning performance, which does not vary significantly over time.

With the BDS-3/GPS UDUC RT-PPP model constructed in this paper, Figs. 10 and 11 exhibit the daily positioning accuracy in 196 days at six stations in the static and kinematic modes, respectively. In the kinematic mode, the daily positioning accuracy is defined as the RMS statistics of positioning errors after full convergence (the results in the first four hours are excluded), while the daily positioning accuracy in the static mode is defined as the RMS statistics of positioning errors for the last half of an hour. Table 3 lists the average and STD statistics of BDS-3/

GPS UDUC RT-PPP positioning accuracy in the east/north/up/3D directions for each station in the static and kinematic modes. According to the average results over six stations, the average positioning accuracy is 4.6, 0.9, 2.5, and 5.6 cm, and 6.4, 3.3, 9.5, and 12.6 cm in the east, north, up and 3D directions in the static and kinematic modes, respectively, which demonstrates that the PPP-B2b UDUC RT-PPP model can meet the requirements for high-precision positioning. Meanwhile, the corresponding STD of positioning accuracy is 1.3, 0.7, 1.7, and 1.5 cm, and 2.3, 1.5, 3.1, and 3.6 cm, respectively. This indicates that the BDS-3/GPS UDUC RT-PPP model based on PPP-B2b service can provide users with stable and reliable high-precision positioning performance.

In order to further evaluate the positioning accuracy of traditional RT-PPP model based on PPP-B2b and the RT-PPP model constructed in this study, Fig. 12 shows the epoch-wise RMS statistics of kinematic positioning errors for the two models. The epoch-wise RMS statistics at an epoch are calculated using the positioning errors at the same epoch from all daily position solutions (using the 24-hour observations from 196 days at six stations). The kinematic positioning errors of traditional model frequently experience large jumps. In contrast, the kinematic

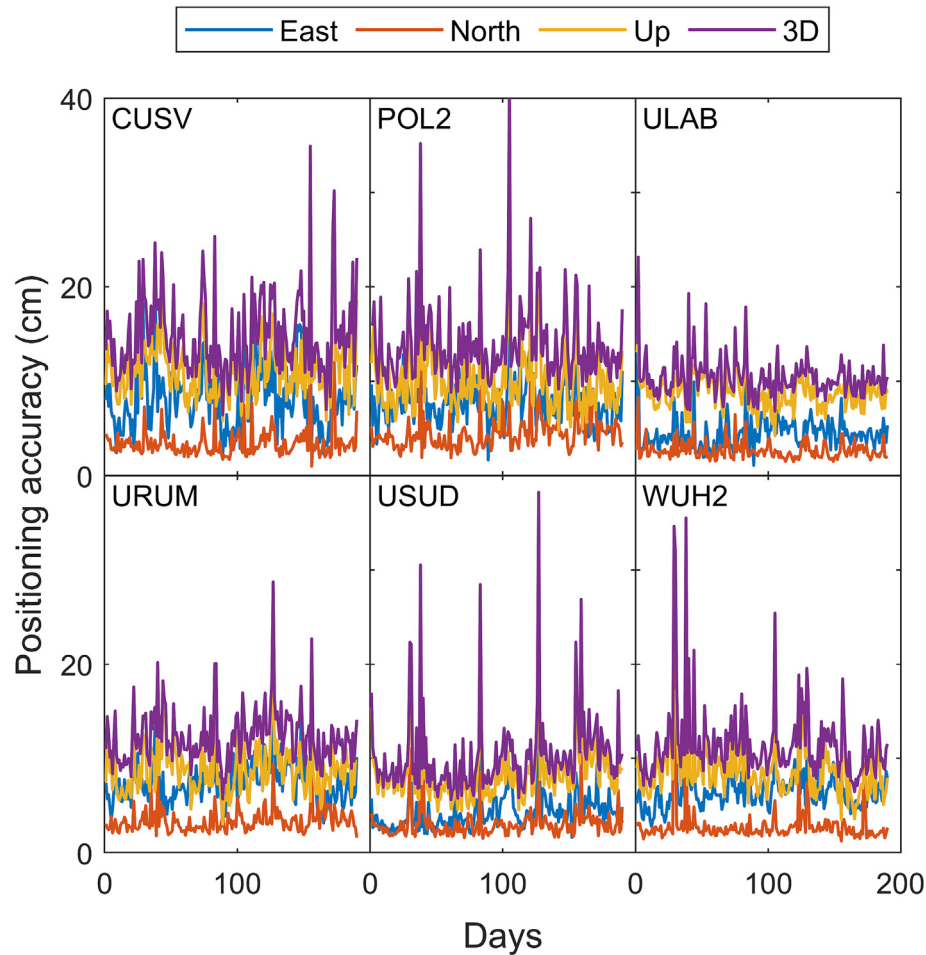


Fig. 11. Daily positioning accuracy in 196 days at six stations in the kinematic mode.

positioning errors of the BDS-3/GPS UDUC RT-PPP model constructed in this study can remain at a relatively low level after a rapid decline during the converging period, indicating that the constructed RT-PPP model can significantly reduce the occurrence of positioning error jumps and ensure the reliability of positioning results.

5. Conclusion

The real-time precise satellite clock offsets of BDS-3 PPP-B2b service comprises three components, namely the datum of satellite clock offsets, the initial systematic bias introduced by initial satellite clock offsets, and the actual satellite clock offsets. The changes in satellite clock datum cause the positioning errors of traditional RT-PPP model to frequently jump after full convergence, leading to unstable and unreliable positioning results. The initial systematic bias reduces the accuracy of code observations, affecting the positioning accuracy in the converging stage and decreasing the convergence speed. This study refines the BDS-3/GPS UDUC RT-PPP model based on the PPP-B2b service. By setting a receiver clock offset parameter for each system to account for the frequent changes in the GPS satellite clock datum, the positioning accuracy

becomes more stable and reliable. Additionally, by setting a compensation parameter for each satellite to mitigate the effects of initial systematic bias, this model accelerates the convergence speed.

The experiments of long-term RT-PPP positioning performance reveal that the average convergence time of the BDS-3/GPS UDUC RT-PPP model based on PPP-B2b service constructed in this study can be 7.5 and 10.2 min in the static and kinematic modes, respectively. Compared to the traditional model, the corresponding convergence time is shortened by 41.3 % and 61.4 %, respectively, indicating that the PPP-B2b UDUC RT-PPP model can achieve the rapid initialization of positioning. Moreover, the average STD of convergence time over 196 days can reach 3.9 and 6.4 min in the static and kinematic modes, respectively. Furthermore, the average positioning accuracies are 4.6, 0.9, 2.5, and 5.6 cm, and 6.4, 3.3, 9.5, and 12.6 cm in the east, north, up and 3D directions in the static and kinematic modes, respectively. This demonstrates that the PPP-B2b UDUC RT-PPP model meets the requirements for high-precision positioning. Additionally, the average STD of positioning accuracy over 196 days is 1.3, 0.7, 1.7, and 1.5 cm, and 2.3, 1.5, 3.1, and 3.6 cm in the four directions in the two modes,

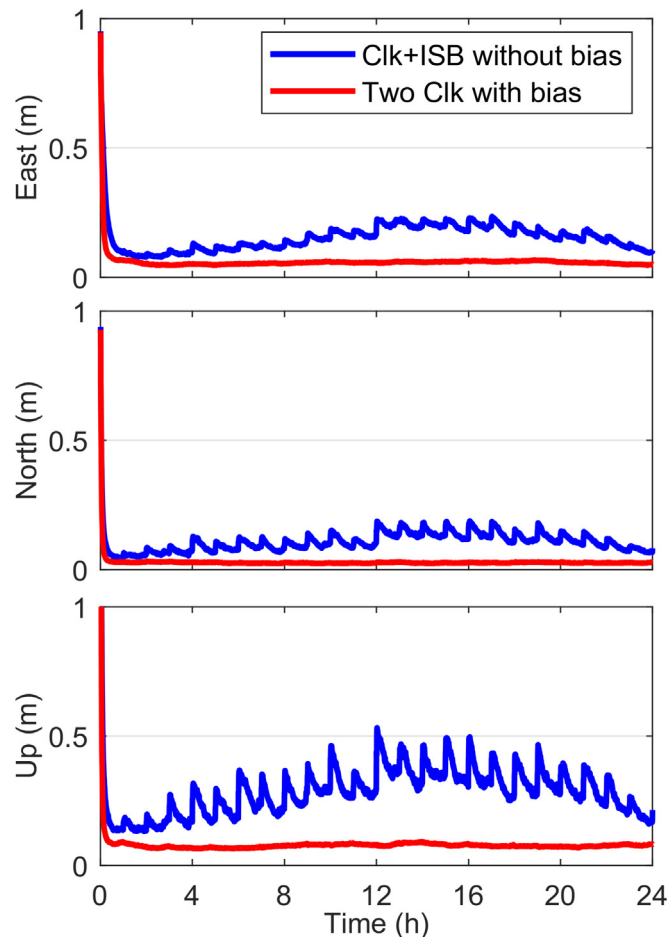


Fig. 12. Epoch-wise RMS statistics of kinematic positioning errors for “Two Clk with bias” and “Clk + ISB without bias” models.

respectively, indicating that the constructed PPP-B2b UDUC RT-PPP model can provide stable and reliable high-precision positioning service.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Ashby, N., 2003. Relativity in the global positioning system. *Living Rev. Relativ.* 6, 1. <https://doi.org/10.12942/lrr-2003-1>.
 Chen, L., Song, W., Yi, W., Shi, C., Lou, Y., Guo, H., 2017. Research on a method of real-time combination of precise GPS clock corrections. *GPS Solut.* 21 (1), 187–195. <https://doi.org/10.1007/s10291-016-0515-3>.

CSNO. 2021. BeiDou Navigation Satellite System Open Service Performance Standard. China Satellite Navigation Office, Beijing, Version 3.0, May 2021. <http://www.beidou.gov.cn/xt/gfzx/202105/P020210526216231136238.pdf>.
 CSNO. 2020. BeiDou Navigation Satellite System Signal In Space Interface Control Document: Precise Point Positioning Service Signal PPP-B2b. China Satellite Navigation Office, Beijing, Version 1.0, July 2020. <http://www.beidou.gov.cn/xt/gfzx/202008/P020230516574071340728.pdf>.
 Dach, R., Bockmann, E. (Eds.), 2024. International GNSS Service Technical Report 2023 (IGS Annual Report). IGS Central Bureau and University of Bern; Bern Open Publishing. doi: 10.48350/191991.
 Ge, Y., Cao, X., Lyu, D., He, Z., Ye, F., Xiao, G., Shen, F., 2023. An investigation of PPP time transfer via BDS-3 PPP-B2b service. *GPS Solut.* 27 (2), 61. <https://doi.org/10.1007/s10291-023-01402-y>.
 Geng, T., Li, Z., Xie, X., Liu, W., Li, Y., Zhao, Q., 2022. Real-time ocean precise point positioning with BDS-3 service signal PPP-B2b. *Measurement* 203. <https://doi.org/10.1016/j.measurement.2022.111911>.
 Guo, J., Geng, J., Zeng, J., Song, X., Defraigne, P., 2024. GPS/Galileo/BDS phase bias stream from Wuhan IGS analysis center for real-time PPP ambiguity resolution. *GPS Solut.* 28 (2), 67. <https://doi.org/10.1007/s10291-023-01610-6>.
 Kouba, J., Héroux, P., 2001. Precise point positioning using IGS orbit and clock products. *GPS Solut.* 5 (2), 12–28. <https://doi.org/10.1007/PL00012883>.
 Malys, S., Jensen, P.A., 1990. Geodetic point positioning with GPS carrier beat phase data from the CASA UNO Experiment. *Geophys. Res. Lett.* 17 (5), 651–654. <https://doi.org/10.1029/GL017i005p00651>.
 Mervart, L., Weber, G. 2011. Real-time Combination of GNSS Orbit and Clock Correction Streams Using a Kalman Filter Approach. In: *Proceedings of ION GNSS 2011*, Portland, OR, USA, 20–23 September 2011, pp 707–711.
 Montenbruck, O., Steigenberger, P., Hauschild, A., 2015. Broadcast versus precise ephemerides: a multi-GNSS perspective. *GPS Solut.* 19 (2), 321–333. <https://doi.org/10.1007/s10291-014-0390-8>.
 Ren, Z., Gong, H., Peng, J., Tang, C., Huang, X., Sun, G., 2021. Performance assessment of real-time precise point positioning using BDS PPP-B2b service signal. *Adv. Space Res.* 68 (8), 3242–3254. <https://doi.org/10.1016/j.asr.2021.06.006>.
 Saastamoinen, J., 1972. Atmospheric Correction for the Troposphere and Stratosphere in Radio Ranging Satellites. In: *The Use of Artificial Satellites for Geodesy*. American Geophysical Union (AGU), pp. 247–251.
 Takasu, T. 2013. RTKLlib: an open source program package for GNSS positioning. <http://www.rtklib.com/>.
 Tang, C., Hu, X., Chen, J., Liu, L., Zhou, S., Guo, R., Li, X., He, F., Liu, J., Yang, J., 2022. Orbit determination, clock estimation and performance evaluation of BDS-3 PPP-B2b service. *J. Geod.* 96 (9), 60. <https://doi.org/10.1007/s00190-022-01642-9>.
 Tao, J., Liu, J., Hu, Z., Zhao, Q., Chen, G., Ju, B., 2021. Initial Assessment of the BDS-3 PPP-B2b RTS compared with the CNES RTS. *GPS Solut.* 25 (4), 131. <https://doi.org/10.1007/s10291-021-01168-1>.
 Wu, J.-T., Wu, S.-C., Hajj, G., Bertiger, W., Lichten, S.M., 1993. Effects of antenna orientation on GPS carrier phase. *Manuscr Geodaet* 18, 91–98. <https://doi.org/10.1007/BF03655303>.
 Wu, P., Lou, Y., Zhang, W., Dousa, J., He, H., Chai, J., Ouyang, Y., Zhang, Z., Zou, X., 2023. Evaluation of real-time kinematic positioning performance of the BDS 3 PPP service on B2b signal. *GPS Solut.* 27 (4), 192. <https://doi.org/10.1007/s10291-023-01532-3>.
 Xiao, P., Sun, F., Wang, K., Xiao, K., Shang, X., Liu, J., 2023. Positioning performance analysis of real-time BDS-3 PPP-B2b/INS tightly coupled integration in urban environments. *Adv. Space Res.* 72 (9), 4008–4020. <https://doi.org/10.1016/j.asr.2023.07.013>.
 Xu, X., Nie, Z., Wang, Z., Zhang, Y., Dong, L., 2023. An improved BDS-3 PPP-B2b positioning approach by estimating signal in space range errors. *GPS Solut.* 27 (3), 110. <https://doi.org/10.1007/s10291-023-01455-z>.

- Xu, Y., Yang, Y., Li, J., 2021. Performance evaluation of BDS-3 PPP-B2b precise point positioning service. *GPS Solut.* 25 (4), 142. <https://doi.org/10.1007/s10291-021-01175-2>.
- Yang, Y., Ding, Q., Gao, W., Li, J., Xu, Y., Sun, B., 2022. Principle and performance of BDSBAS and PPP-B2b of BDS-3. *Satell. Navig.* 3 (1), 5. <https://doi.org/10.1186/s43020-022-00066-2>.
- Yao, Y., He, Y., Yi, W., Song, W., Cao, C., Chen, M., 2017. Method for evaluating real-time GNSS satellite clock offset products. *GPS Solut.* 21 (4), 1417–1425. <https://doi.org/10.1007/s10291-017-0619-4>.
- Zang, J., Fan, S., Xu, C., Li, Z., Fang, R., Lou, Y., 2024. Performance assessment of the BDS-3 PPP-B2b service for real-time earthquake source description: a case study for the 2021 Mw 7.4 Maduo earthquake. *GPS Solut.* 28 (1), 26. <https://doi.org/10.1007/s10291-023-01570-x>.
- Zhang, W., Lou, Y., Song, W., Sun, W., Zou, X., Gong, X., 2022. Initial assessment of BDS-3 precise point positioning service on GEO B2b signal. *Adv. Space Res.* 69 (1), 690–700. <https://doi.org/10.1016/j.asr.2021.09.006>.
- Zhang, X., Li, X., Guo, F., 2011. Satellite clock estimation at 1 Hz for realtime kinematic PPP applications. *GPS Solut.* 15 (4), 315–324. <https://doi.org/10.1007/s10291-010-0191-7>.
- Zhou, H., Fu, W., Wang, L., Li, T., Wu, Y., Chen, R., Li, J., 2023. Multi-frequency BDS-3 real-time positioning performance assessment using new PPP-B2b augmentation service. *IEEE Sens. J.* 23 (5), 4994–5002. <https://doi.org/10.1109/JSEN.2023.3235901>.
- Zumberge, J.F., Heflin, M.B., Jefferson, D.C., Watkins, M.M., Webb, F. H., 1997. Precise point positioning for the efficient and robust analysis of GPS data from large networks. *J. Geophys. Res. Solid Earth* 102 (B3), 5005–5017. <https://doi.org/10.1029/96JB03860>.